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Multi-Stage Financing of High-Potential Ventures

Venture capital returns are driven by "hits". This is because the vast majority of VC investments fail entirely and a handful of startups account for most of the funds' returns. **Exhibit 1** shows the distribution of returns for a single large and successful venture capital firm that has invested more than $1 billion over the last decade. Panel A of **Exhibit 1** shows that about 60% of the investments were terminated at a loss, and together accounted for under 5% of the gross returns. On the other hand, 5% of the most successful investments returned over 10 times the capital invested and accounted for over 50% of the gross returns of the fund.[1] Conditional on making an investment, however, Panel B of **Exhibit 2** shows the limited ability of the venture capital investors to tell the most successful investments from the less successful ones—even for these experienced investors. It documents the scores that partners assigned to startups at the time of first investment and compares them to the ultimate outcome of the startups. Panel B highlights that the distribution of scores assigned to startups that ended up performing extremely well is statistically no different from the distribution of scores assigned to investments that ultimately failed or had mediocre returns.

As seen in **Exhibit 1**, most startups VCs invest in do not achieve their expected outcome. The difficulty in assessing the potential of a startup prior to an investment means that VCs also sometimes pass on deals that end up becoming extremely successful. For example, Google was in many ways a long-shot investment when it was founded in 1999. Its founders were students, it was entering a space with many established competitors and it had no clear business model. In fact, more than one venture capitalist had turned down the opportunity to invest in Google at the time. One such firm was Bessemer Ventures, whose partner David Cowan, on being asked to meet with the two Google founders working in the garage, is believed to have quipped, "Students? A new search engine? ... How can I get out of this house without going anywhere near your garage?" Not everyone is as candid as about great deals they passed on as Bessemer Ventures, who have been so successful that they can playfully mock at their "anti-portfolio" of investments.[2] However, Fred Wilson of Union Square Ventures has similarly written about his regret at passing on Airbnb, a startup that lets people rent rooms or homes to prospective guests online. Started in 2008, Airbnb listed 500,000 properties in more than 34,000 cities across 192 countries (as of December, 2013), far larger than any hotel chain—thereby making it the largest lodging company and brand in the world (Levere, 2013). Wilson (2011) writes of Union Square Venture's decision to pass on the deal in 2009 that "we couldn't

[1] This distribution is consistent with the returns across all venture capital investments in the United States from 1985-2009. Kerr, Nanda and Rhodes-Kropf (2014) estimate that 55 percent of US startups that received venture capital from 1985-2009 were terminated at a loss, while 6 percent accounted for 50 percent of the gross return generated over the period.

[2] Bessemer Venture Partners, "Anti-Portfolio," 2015, http://www.bvp.com/portfolio/antiportfolio.

Professor Ramana Nanda prepared this note as the basis for class discussion.

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wrap our heads around air mattresses on the living room floors as the next hotel room…. Others saw the amazing team that we saw, funded them, and the rest is history."[3]

These examples highlight the difficulty that VCs have in evaluating investment opportunities, driven by the fundamental uncertainty facing high-potential ventures at their earliest stages.[4] In this context of extreme uncertainty, multi-stage financing allows investors to learn about a venture's potential over time, without committing the full amount upfront. The abandonment option that arises from the ability to stop investing when intermediate information is negative is particularly valuable for venture capital investors, because most new ventures fail completely, and only a few go on to become extremely successful. The process of experimentation, where investors learn about the viability of a radical new idea through an initial investment, interpret intermediate results, and decide whether to continue or abandon their investment, is therefore a key aspect of Entrepreneurial Finance.

This note introduces a framework for understanding multi-stage financing and abandonment options in venture capital. Section 1 provides an introduction to the venture capital investment model and maps the decision tree framework to the VC method for valuing new ventures. This provides an intuitive way to understand why startups in some industries have higher initial valuations and larger step-ups across rounds of funding, as well as why valuations might be expected to vary across time. It also helps to map the techniques around 'lean startups' to the financing and valuations of startup ventures. Section 2 describes a number of challenges and constraints to experimentation by investors, and the implications this has for the financing strategies of startup ventures. Finally, Section 3 outlines how the cost of experimentation has fallen significantly in certain sectors over the prior two decades and discusses some implications for multi-stage financing.

1. Mapping Startup Valuations to a Decision Tree Framework

As noted above, venture capital financing occurs in stages rather than the full investment being provided upfront (as is the case in settings where there is no uncertainty about the potential outcome). A key element of staged financing is the need to set appropriate milestones at every round of funding. These milestones are specific achievements that significantly increase the probability of the venture's success (or the value conditional on success) and hence justify another round of financing. They could entail receiving a patent on the technology, getting a first customer, demonstrating "traction" with end users, passing regulatory hurdles such as FDA trials, etc. Since staged financing is tied to achieving milestones that create information about the future prospects of the venture, each round of funding can be seen as an experiment that generates information about the venture's probability of success and its value conditional on that success. Experiments that meet their milestones therefore generate positive information and increase the value of the company, allowing the entrepreneur to seek the next round of funding without giving up as much equity. On the other hand, experiments that reveal negative information about the prospects of the startup allow investors to abandon their investment without investing the full amount, (or exercise their control rights to replace the CEO or push for a different commercialization strategy if the venture does not continue on track to being successful). This abandonment option is particularly useful for VCs, as it

[3] Source: Wilson, Fred, 2011, "Airbnb." http://www.avc.com/a_vc/2011/03/airbnb.html

[4] The use of the term "High-Potential" ventures captures the notion that startups commercializing new technologies, engaged in business model innovations or other forms of disruption are significantly different from the majority of new startups, most of which enter with existing business models, proven technologies and where often the founder has no ambition to grow the business aggressively.

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allows investors to 'cut their losses early' and deploy the capital towards another investment in the portfolio (or to make a new investment in the hope that it has a better outcome).

The following examples use a decision tree framework to make these ideas concrete: Consider a startup that takes $12 million to commercialize and has a 10% chance of being worth $ 100 million, and a 90% chance of being worth zero. As seen in Panel A of **Exhibit 2**, This startup will not be financed, because its expected value is negative (10% × 100 M - $12 M = - $2 M). But imagine that investors can break this $12 million into two stages. They can first invest $2 million in an experiment that will reveal if the startup has either a 50% chance of working (positive intermediate information), or is guaranteed to fail (negative intermediate information), before deciding whether to invest the remaining $10 million. This allows the investor to abandon their investment if intermediate information is negative. Suppose further that the probability of the experiment having positive intermediate information is 20%, so that in this example the 10% chance of overall success is unchanged. As shown in Panel B of **Exhibit 2**, we can see that the experiment now makes the investment profitable. The expected value of the investment if the experiment is successful is $ 40 million (50% × $100 M - $ 10 M), which means that (since the probability of the experiment working is 20%), any experiment that costs up to $ 8 million will be profitable. Thus even though the initial investment of $12 million was not profitable, breaking the investment into stages (where the investor does not have to continue if intermediate information is negative) allows an investment up to $18 million in total! The value of the abandonment option in this example is thus $6 million (the expected value with the option to abandon) minus -$2 million (the expected value without the option to abandon) = $8 million. This is the power of experimentation.

The value of experimentation stems from two sources. The first is the cost of the experiment. Not surprisingly, holding all else equal, experiments that cost less are more valuable. The second source of value is the amount of information an experiment generates. The amount of information an experiment generates is defined as the difference in the expected value of the startup when intermediate information is positive compared to when intermediate information is negative. In the example above, this was $40 million – 0 = $40 million. The most informative experiments can perfectly inform an investor about the probability of success. That is, in a perfect experiment, the probability of success is 100% (the startup was guaranteed to be successful) if intermediate information is good, while the probability of success is zero (the startup is guaranteed to go bankrupt) if intermediate information is negative. **Exhibit 3** shows that when using the same values as the example above, the most informative experiment would yield a difference in expected value of $100 million. On the other hand, a completely uninformative experiment is one where the expected value of the startup when intermediate information is positive is the same as that when intermediate information is negative. In the example above, suppose the probability of success was 10% when the experiment was successful, and was also 10% when the experiment failed. In this case, the expected value of the second stage was the same (10% × $100 M - $ 10 M = 0) regardless of whether the experiment failed or was successful, and hence was completely informative. In fact, it can be seen that the unconditional probability of success here is also 10% (20% × 10% + 80% × 10%), but since the experiment was completely uninformative, it added *no value*. Hence the expected value of this investment is the same (- $2 million) as the case where there was no experiment involved (Panel A of **Exhibit 2**).

We can see from this example that experimentation is valuable when it allows an investor to change their actions in the second period – that is, to abandon their investment when intermediate information is negative. The more informative an experiment is, the less likely an investor is to deploy capital in the second stage towards a project that will fail. Informative, or discriminating investments, are thus ones where the value of the *abandonment option* is high.

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We now take these insights on the drivers of value for experiments and map them to capitalization tables associated with venture capital investments. **Exhibit 4** uses the same example seen in Panel B of **Exhibit 2**. As can be seen from **Exhibit 4,** the startup has a 50% chance of being successful in stage 2. If successful, its value will be $100 million. If it fails, the startup will be worth zero. Since it costs $10 million to scale this venture at the second stage, the expected value conditional on the experiment working is $ 40 million (50% × $100 M - $ 10 M). Further, since the experiment has a 20% probability of yielding positive intermediate information and the investment required to run the experiment is $2 million, the expected value prior to an investment is 20% × $40 M - $ 2 M = $6 million. The expected value at each of these nodes of the decision tree is equivalent to the pre-money valuation of the startup at that round. Given that we know the investment that is made at each stage, this allows us to calculate the post money valuation and the percentage ownership at each round of funding. In the first round, the "early stage investor"(VC1) takes 2/(2+6)=25%, leaving founders with 75% of the firm's equity. If the experiment yields positive information, the "late stage investor" (VC2) takes 10/(10+40)=20% in the second round.[5] If VC2 invests, VC1's ownership is diluted by 20%, so that VC1 now owns 25%*80% = 20% and hence the final ownership is such that both VCs each own 20% and the founder owns 60%. If the second stage is also successful, then the $100 million value is divided between the owners in proportion to their ownership.

Assuming that the second round of funding was raised 2 years after the first investment and the startup achieved success in 5 years, Exhibit 4 shows that VC1 generates an IRR of ~ 59% ($2 \times (1.59)^5 = 20$) and VC2 generates an IRR of ~ 25% ($10 \times (1.25)^3 = 20$) if the startup is successful.[6] This example highlights how the venture capital method of valuation maps onto the decision tree framework in Exhibit 4. Venture capital investors often apply a high IRR to a "success scenario" as a way to capture the low probability of success for startups.[7] Exhibit 4 shows the underlying mechanics for such a logic. VC2 faces an investment with a higher probability of success than VC1 because VC2 will only invest if the experiment is a success. The success of the initial experiment will have increased the startup's ultimate probability of success from 10% at the early stage to 50% at the growth stage. On the other hand, VC1 faces a 90% likelihood of losing $2 million (80% chance of the experiment failing and 50% chance of scaling not working, even if the experiment works). Thus, VC1 needs to be compensated for this lower likelihood of success, receiving a higher return if the venture is successful. This mapping is useful in that it highlights how a negotiation between entrepreneurs and investors on valuation is effectively a negotiation around the implicit probability of failure being assigned by the investors in such a decision tree framework. Factors that lower the perceived probability of failure in a given or future round (e.g. a stronger team, a path-breaking technology) also lead the investor to require a lower IRR in the successful state, and hence will lead to a higher valuation.

Having mapped the general notion of success probabilities in decision trees to IRRs used by venture capital investors, we turn to specifically examining how experimentation impacts the ownership shares of the founder and the various investors. It is worth noting that while the focus in the examples below will be on factors of initial experiments that impact valuation, there are of course

[5] Note that there is nothing that requires there to be two investors. The same VC (or both VCs) could invest across both rounds. This notation makes it clearer to see the drivers of value and dilution.

[6] The note also assumes away systematic risk and sets the risk free rate equal to zero. Throughout this note, the assumption is that VCs compete away all rents, so that the NPV of their investment is always zero. The expected value of an investment at each stage will therefore always equal the investment itself, although *conditional on success*, the investors will have high IRRs.

[7] For more details, see "A Method For Valuing High-Risk, Long-Term Investments: The 'Venture Capital Method'" HBS Technical Note 288-006.

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other factors that impact initial valuations. Looking at **Exhibit 4**, for example, a higher expected value if successful (e.g. $200 million rather than $100), a lower overall investment (e.g. $ 5 million instead of $12 million) and a higher unconditional probability of success (e.g. 20% instead of 10%) will all make the investment much more attractive and reduce the dilution in the first round. Note, however that these factors equally impact all-or-nothing bets as well. That is, a higher ultimate value, a lower cost and a greater probability of success also make the startup in Panel A of **Exhibit 2** more valuable. This does not discount their importance – entrepreneurs need to highlight how valuable their startup will be if it achieves its potential, that it is capital light (perhaps due to an attractive business model) and that the probability of success is high (because of the team, the opportunity, the technology, the deals with investors and other constituents, etc.). [8] Each of these factors makes all startups more valuable, irrespective of the order in which information is revealed or the degree to which the investment is front-loaded vs. disproportionately during the scale-up period. This note aims to emphasize how the attributes of experimentation across multiple rounds of funding have an impact on the ownership for entrepreneurs and for early-stage and later-stage investors. We turn to a discussion of these below.

As noted above, experimentation generates value from two sources: the cost of the experiment and the information generated by it. Looking first at the cost of experimentation, consider a startup with the same probabilities of success across the rounds and the same value if successful as in **Exhibit 4**, but where the initial experiment costs $7 million instead of $2 million. The decision tree and capitalization table for this startup is shown in **Exhibit 5**. In order to make it comparable to the startup in **Exhibit 4**, we assume that the total investment required to commercialize the startup is the same (otherwise the startup would be an unambiguously worse investment in addition to having a more expensive experiment), hence the capital required to scale this startup is assumed to be $5 million. We can see from Exhibit 5 that now the expected value in the second stage conditional on the experiment working is $45 million and the expected value prior to an investment is $2 million. In this example, VC1 takes 7/9= 78% of the startup's equity to fund the early stage experiment, and VC2 takes 10%, diluting VC1's stake to 70% at the completion of the second round of funding. If the startup is successful after the second round of funding, VC1 will get $70 million, VC2 will get $10 million and the entrepreneur will get $20 million. Comparing the capitalization tables in **Exhibit 4** and **Exhibit 5**, one can see that the higher cost of the early experiment makes the experiment less valuable for VC1 *even though the total funding required for both startups is the same*. VC1 therefore has to take a 78% share of the startup to be willing to finance the experiment (compared to only 25% if the initial experiment had cost $2 million).

Exhibit 6 documents the capitalization table for a startup where the experiment is *less informative*. The early stage experiment for this startup is less informative because the expected value conditional on the experiment working minus the expected value conditional on the experiment failing is $10 million, compared to $40 million in **Exhibit 4**. Note however that the startup in this example has an experiment with a *greater* chance of being successful (50% compared to 20%). This is because we want to compare two startups with the same unconditional probability of success but with different levels of information generated by the initial experiment. If we kept the probability of the experiment working the same as in **Exhibit 4**, this would lead to both a worse project (a lower unconditional value of success and one with a less valuable experiment).

Exhibit 6 documents that VC1 needs to take a 40% share of the startup to finance the early stage experiment. VC2 needs to take a 50% investment, diluting VC1's share down to 20%. If the startup is successful after the second stage financing, VC1 will get $20 million and VC2 will get $50 million. Comparing Exhibit 4 to Exhibit 6 is instructive. First, **Exhibit 6** shows that VC2 needs to take a larger

[8] For example, see "Some Thoughts on Business Plans", HBS Background Note 897-101.

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share of the startup to finance the scale up. This is because the experiment has yielded less information about whether the startup will ultimately be successful, making the experiment less valuable for VC2. VC2 faces a lower likelihood of successfully scaling the startup. Conditional on success, therefore, VC2 has a 71% IRR compared to the 25% in **Exhibit 4**. Anticipating the larger share that VC2 needs to take if the experiment is successful, VC1 factors in the expected dilution and has to take a larger share to finance the experiment. Importantly the dollar return, and hence the IRR, that VC1 gets if the firm is successful is identical to that in **Exhibit 4**. That is, a less discriminating experiment means that VC1 needs to take a larger share of the startup *even though the unconditional probability of success is the same*. Put differently, the degree to which information is revealed early vs. late has a bearing on valuations.

Exhibits 4-6 show that the step-up in value (which is the pre-money valuation in round 2 divided by the post money valuation in the first round) is inversely proportional to the probability of the experiment working. In Exhibit 6, the probability of the experiment working was 50%, and the step up in value was 10/(2+3)=2X which is the same as 1/50%. In **Exhibit 4**, the probability of the experiment working was 20%, the step up in value was 40/(7+1)= 5X which is the same as 1/20%. Thus, more discriminating experiments lead to higher step-ups in value. This is intuitive, because it means that more uncertainty has been reduced when the experiment actually works. Although the initial experiment may be less likely to work, when it does, it significantly increases the probability of success (50% in **Exhibit 4** compared to 20% in **Exhibit 6**). It can be seen therefore that startups (or sectors) in which the initial experiments are less informative will lead to lower step-ups. Anticipating these, early investors will take large shares in early rounds, leading to much lower valuations for entrepreneurs.

The examples above provide some general lessons for entrepreneurs financing their high-potential ventures. First, experimentation allows investors and entrepreneurs to start firms that could not have been started in an all-or-nothing bet because the ability for investors to abandon an investment when intermediate information is negative allows them to take a lower share of the startup in the successful states of the world. While entrepreneurs do not like investors to exercise abandonment options, this also highlights that the ability to exercise abandonment options sometimes allows investors to finance an initial experiment that they may have been unwilling to fund otherwise!

Second, the high IRRs that VCs receive when startups perform well are best seen as a reflection of the low unconditional probabilities of success for the startups they invest in. The high failure rates faced by such investors imply that they need to own a high enough share of the startup in the successful states to compensate for the losses when the startup fails. Therefore, when the startups are successful, investors are seen to experience high IRRs.[9]

Third, since the expected value is equivalent to the pre-money valuation of the round, all things being equal, startups with more expensive and less discriminating early experiments will have lower initial valuations and suffer more dilution. This explains why industries where initial experiments are expensive and not as discriminating (e.g., biotech or clean energy) will be ones in which startups have lower initial valuations and lower step-ups across rounds of funding. Founders will be left with a lower share the firm, so that even if the ultimate exit is the same as a successful consumer internet

[9] Note that in our examples, the unconditional probability of success multiplied by the dollar return conditional on success is always equal to the investment. This is because we have assumed that entrepreneurs have choice, so that the VC market is competitive and they earn NPV=0. Note also that we have focused on the aspect of decision trees that relate to the option to abandon a project and have assumed a risk free rate that is set equal to zero. Calculating the value of options in the presence of beta risk is not covered in this note. A good resource for this topic is Metrick, 2007, "Venture Capital and the Finance of Innovation".

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startup, the dollar value of their return will be less. This also highlights why the "minimum viable product" in the lean startup model is such a powerful tool for entrepreneurs; it forces entrepreneurs to design cheaper and more discriminating experiments that reduce dilution substantially and lead to a higher return for founders and the management team.

2. Constraints to Experimentation in Multi-Stage Financing

Decision trees provide a useful framework for understanding the drivers of value across multi-stage financing, and the steps that entrepreneurs can take to reduce the dilution they face. However, the real world is inherently more complicated than the decision trees in the attached Exhibits. Often, information that comes back from early experiments is mixed – ventures have met some milestones but not others, the startup has had some successes and some setbacks, yet investors need to make binary decisions based on incomplete information about whether to continue funding a startup or whether to shut it down. The specific incentives facing individual investors and contextual factors governing the financing landscape at the time can therefore have important interactions with the results of experiments. This section highlights some of the important interactions and discusses implications for entrepreneurs.

2.1 Constraints to Exercising Abandonment Options

Thus far we have considered abandonment options from the perspective of investors. However, entrepreneurs are at the receiving end of these options being exercised, either in the form of the startup not receiving follow-on funding, or in terms of founders being replaced. When entrepreneurs have a choice between investors, they will prefer investors with a reputation for being "founder friendly" or "not so quick on the trigger". Investors therefore often like to build a reputation for being founder friendly, which in the context of the decision tree framework is consistent with not exercising abandonment options even when intermediate information is negative and the expected value of pursuing a path is negative. That is, the value they place on building a reputation for being founder friendly (and being able to attract the best deal flow/talent) outweighs the costs associated forgoing abandonment options. From the investor's perspective, this implies that they may choose to back startups where abandonment options are not as valuable. From an entrepreneur's standpoint, it implies that they should be willing to take a lower valuation from an investor who has a founder friendly reputation (since the investor needs to be compensated when things go well in return for not being as quick on the trigger when things go badly).

2.2 Continuation and Financing Risk

The venture capital industry is notorious for cycles in the availability of capital. This can impose substantial costs on high-potential ventures, as the process of staging means that startups return to the capital markets more frequently and hence are more vulnerable to the ebbs and flows in the availability of capital. Venture capital investors refer to this as financing risk – the risk that shocks to the availability of capital lead otherwise healthy firms to go bankrupt because they were unable to raise follow-on funding for their ventures. In the conceptual framework of decision trees, financing risk is equivalent to a lower probability of the first stage experiment working, and hence lowers the value of the abandonment option. Times when financing risk is high (that is, when investors feel they are less likely to get follow-on funding) will be ones with lower valuations, as it is equivalent to a lower probability of the experiment working.

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3. Falling Costs of Initial Experiments

Technological changes in the last few decades have led to substantial falls in the cost to learn about the potential of new ventures. For example, Mark Andreessen notes that "In the '90s, if you wanted to build an Internet company, you needed to buy Sun servers, Cisco networking gear, Oracle databases, and EMC storage systems... and those companies would charge you a ton of money even just to get up and running. The new startups today, they don't buy any of that stuff... They're paying somewhere between 100x and 1000x [less] per unit of compute, per unit of storage, per unit of networking''. Andreessen also notes that the rise of services such as Amazon's Elastic Compute Cloud (EC2) have transformed many infrastructure costs from upfront capital expenditures to subscription services that can scale with a company as it grows.[10]

In the context of the decision tree framework, it can be seen at a falling cost of early experiments allows startups that are 'long shot bets' to get funding. For example, looking at **Exhibit 4**, when the cost of the initial experiment is $2 million, any startup whose initial experiment has a lower than 5% likelihood of working will not get funded. However, if the cost of the initial experiment fell to $1 million, startups whose likelihood of the initial experiment working is between 2.5% and 5% will now become profitable investments. While no investor knows the exact probability of initial experiments working, this example demonstrates how the dramatic fall in the cost of early experiments (at times as much as 1000X) can impact the types of startups that will get an initial investment. For example, less proven entrepreneurs are now more likely to get financing, democratizing access to funding.

The falling cost of early stage experiments has also allowed for a proliferation of funding sources. The rise of angels, super-angels and crowdfunding platforms can be seen as a response to these technological shifts in the cost of starting new ventures. However, the decision tree framework also suggests caution for investors and regulators: while the the falling cost of experimentation has made 'long shot bets' viable investments, they still are long shot bets. That is, the most likely outcome for these startups is failure. Thus, it should not be surprising that a smaller share of these ventures receive follow-on venture capital financing (e.g. the "Series A crunch") and the proportion of startups that receive funding should be *expected* to have a higher rate of failure.

[10] Source: Douglas Macmillan, "Andreessen: Bubble Believers 'Don't Know What They're Talking About'," Wall Street Journal, January 3, 2014.

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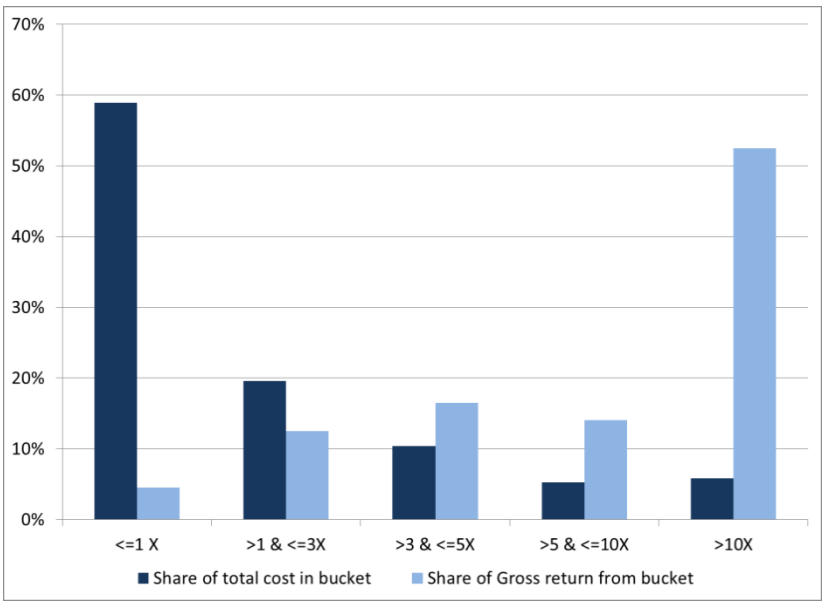
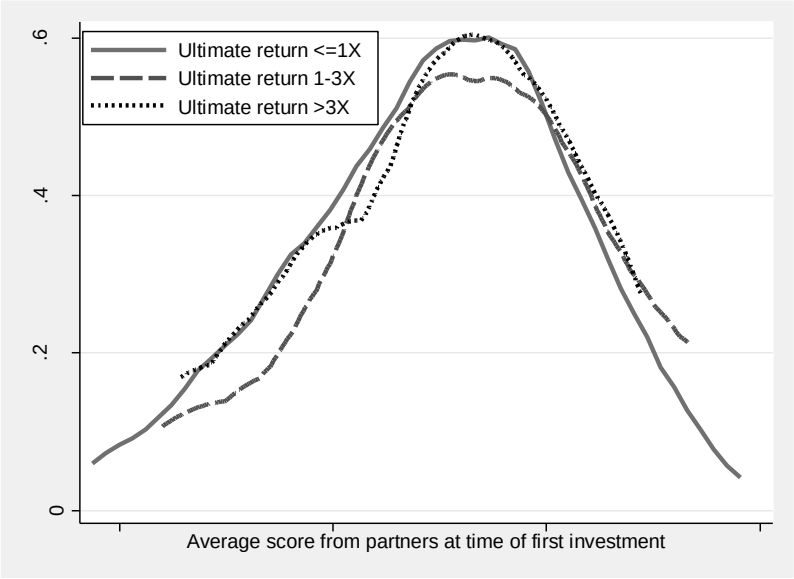
Exhibit 1 Return Profile of Venture Capital Investments

Panel A: Distribution of total cost and gross return for a venture capital firm



Panel B: Distribution of scores assigned at time of first investment, by outcome



Source: Kerr, Nanda and Rhodes-Kropf (2014), Entrepreneurship as Experimentation. Journal of Economic Perspectives 28, no. 3 (Summer 2014): 25–48. [Copyright American Economic Association; reproduced with permission of the Journal of Economic Perspectives]

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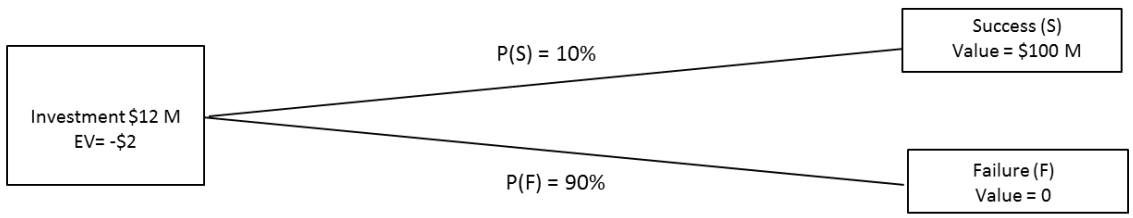
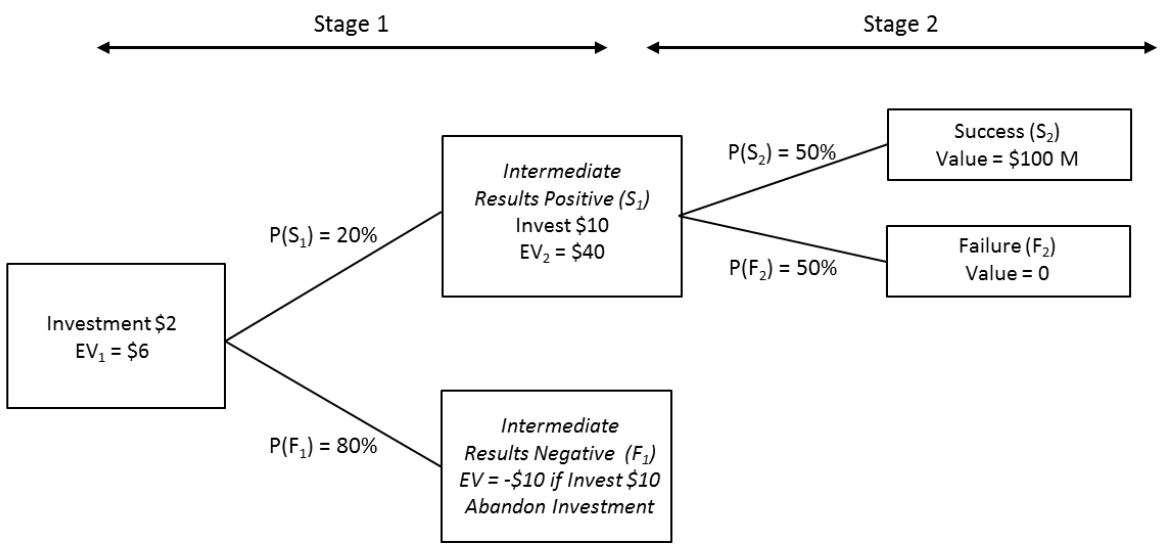
Exhibit 2 Value of Staged Financing

PANEL A: All-OR-NOTHING BET



Expected Value (EV) = (10% × $100) + (90% × 0) - $12 = -$2 M

PANEL B: STAGE FINANCING WITH OPTION TO ABANDON IF INTERMEDIATE RESULTS ARE NEGATIVE



Expected Value of Stage 2 if Intermediate Results Positive: (EV_2) = (50% × $100) - $10 = $40 M

Expected Value of Stage 1 (EV_1) = (20% × $40) + (80% × 0) - $2 = $6 M

Source: Casewriter calculations.

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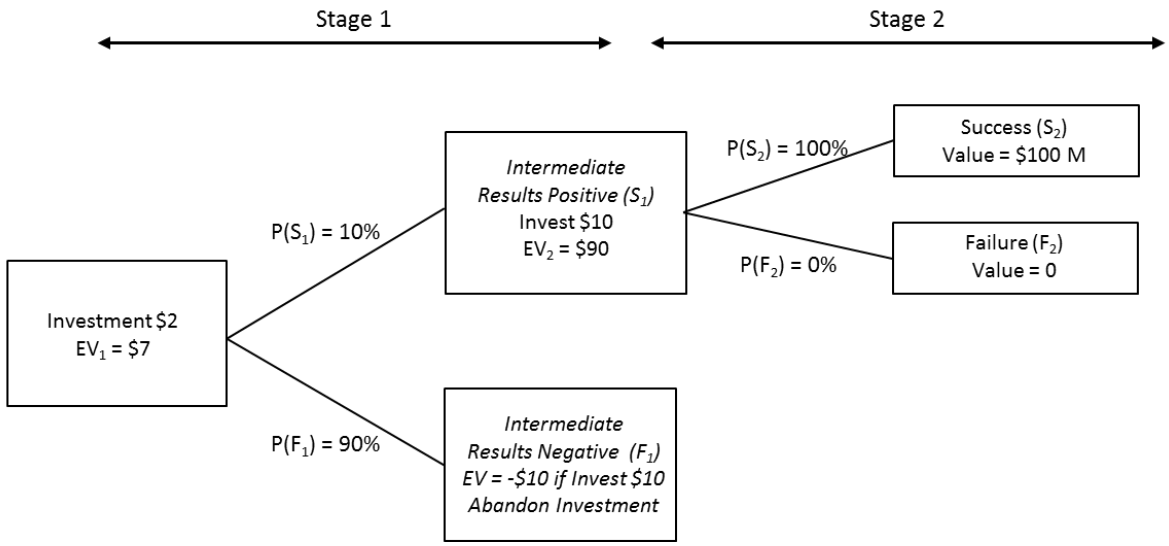
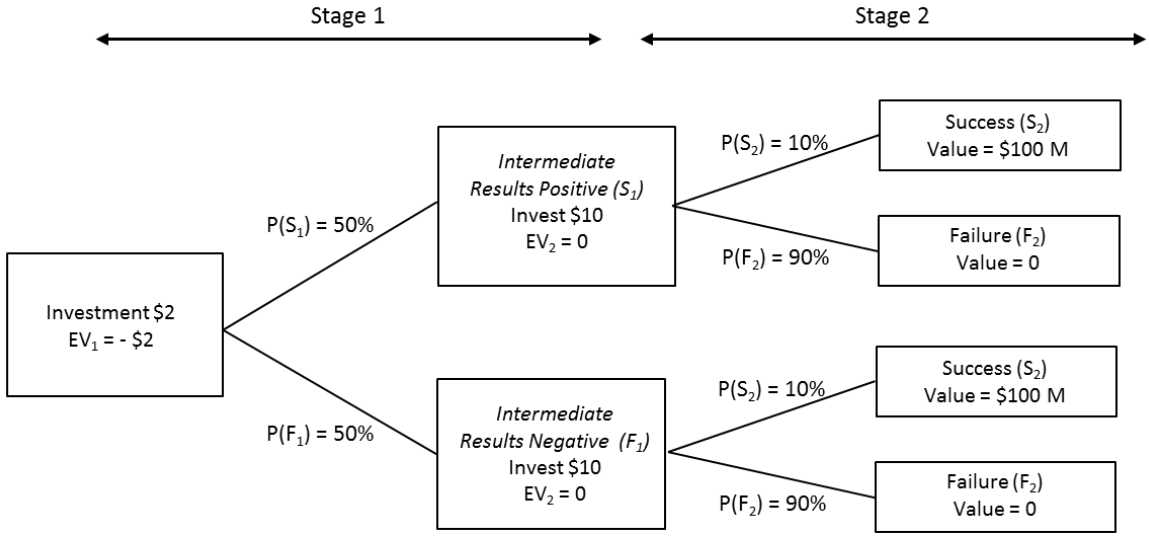
Exhibit 3 Information generated by First Stage Experiment

PANEL A: PERFECTLY INFORMATIVE FIRST STAGE EXPERIMENT



PANEL B: UNINFORMATIVE FIRST STAGE EXPERIMENT



Source: Casewriter calculations.

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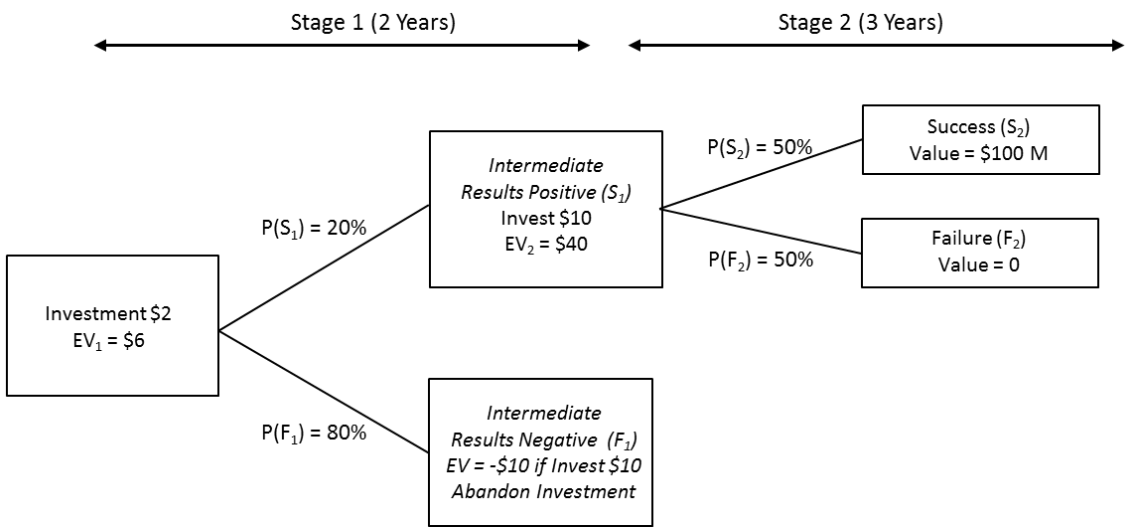
Exhibit 4 Mapping Decision Tree Framework to Capitalization Tables

PANEL A: DECISION TREE



PANEL B: CAPITALIZATION TABLE

	Ownership after Stage 1	Ownership if VC2 invests at Stage 2	$ Return if successful after Stage 2	IRR if successful after Stage 2
Entrepreneur	75%	60%	$ 60 M	
VC 1	25%	20%	$ 20 M	58%
VC 2		20%	$ 20 M	26%
Total	100%	100%	$ 100 M	

Source: Casewriter calculations.

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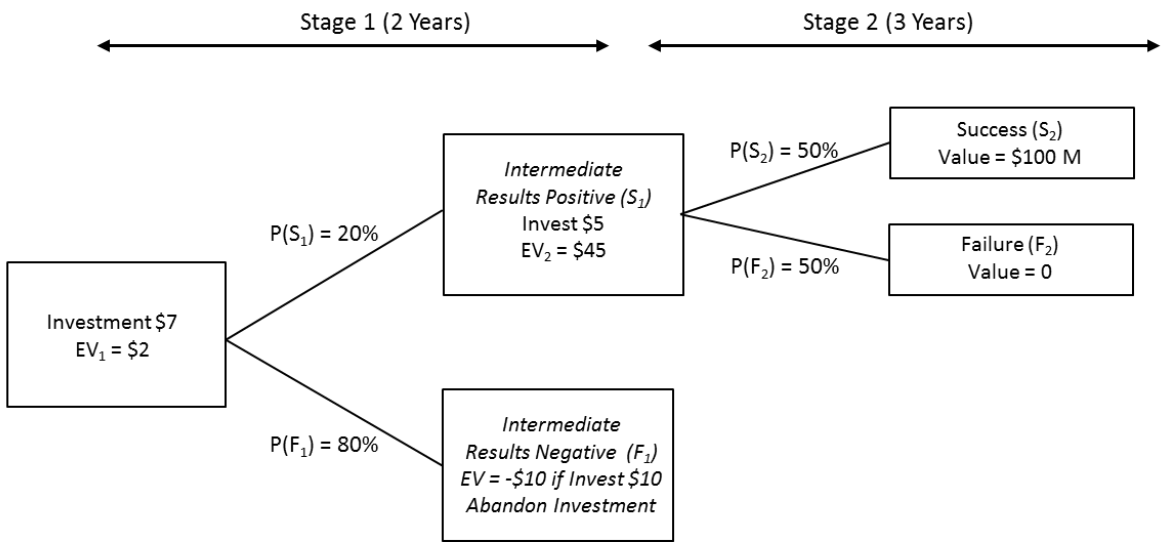
Exhibit 5 Higher Cost of Initial Experiment

PANEL A: DECISION TREE



PANEL B: CAPITALIZATION TABLE

	Ownership after Stage 1	Ownership if VC2 invests at Stage 2	$ Return if successful after Stage 2	IRR if successful after Stage 2
Entrepreneur	22%	20%	$ 20 M	
VC 1	78%	70%	$ 70 M	58%
VC 2		10%	$ 10 M	26%
Total	100%	100%	$ 100 M	

Source: Casewriter calculations.

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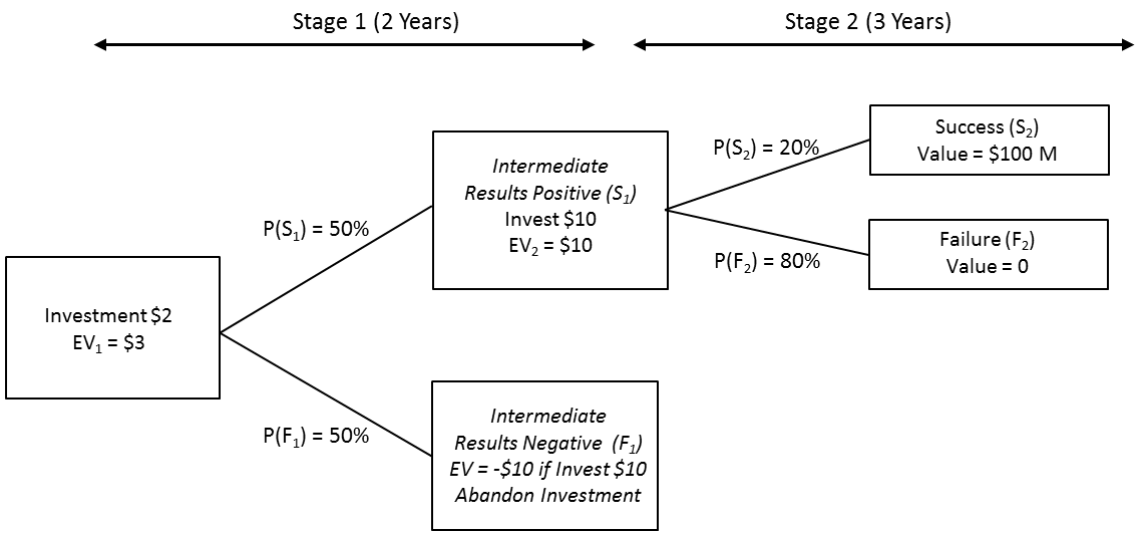
Exhibit 6 Less Informative Initial Experiment

PANEL A: DECISION TREE



PANEL B: CAPITALIZATION TABLE

	Ownership after Stage 1	Ownership if VC2 invests at Stage 2	$ Return if successful after Stage 2	IRR if successful after Stage 2
Entrepreneur	60%	30%	$ 30 M	
VC 1	40%	20%	$ 20 M	58%
VC 2		50%	$ 50 M	71%
Total	100%	100%	$ 100 M	

Source: Casewriter calculations.

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